



Bats live the longest

Adjusting for their size, there are 19 mammals that live longer than humans. 18 of them are bats. <https://digit.in/july19-51>



Inca Canals

1,400 year old ancient Inca canals could boost the dwindling water reserves in coastal Peru. <https://digit.in/july19-48>

the incidence and geographic range of the virus were exponentially increased. The outbreak of SARS and recurrent Ebola crisis in Central Africa accentuate the ability of new and existing diseases becoming significant international health threats. Additionally, the human ability to combat infectious diseases is also constantly challenged by the emergence of pathogen drug resistance.

Therefore, despite the decline of deaths caused by infectious diseases, the complexity of our interactions with disease-causing agents are very significant, even now. These diseases are still a major cause of mortality across the globe and are actually responsible for a quarter to a third of all deaths around the world. The harrowing factor? Most of these deaths are principally unavoidable.

OUR BODIES' DEFENCES AGAINST PATHOGENS

Humans constantly come in contact with disease-carrying pathogens in the form of viruses, bacteria and parasites. Fortunately, the largest organ in our body - our skin along with the specialised lining of our guts and other parts of our body that are exposed to the environment prevent these intruders from entering. The body's defences live up every time a pathogen breaches these barriers. The defence that comes into play if the external elements fail, is called the human immune system which has two branches - innate and adaptive.

The innate immune system is basically the first line of defence. It comprises of special molecules that identify 'fingerprints' or patterns in proteins or genetic material present in pathogens. These molecules alert the innate system about the presence of a pathogen, which then produces cytokines, also known as effector molecules. The effector molecules kick off the process of eliminating pathogens. The adaptive immune response doesn't kick in until much later. When it does, it must assemble a much more robust defence and destroy cells which are infected with a virus. They can even neutralise the virus or toxins produced by these bacterial pathogens.

Additionally, the adaptive system has a memory and activates extremely fast if the same pathogen invades our system again. Both systems, however, are linked and the innate system prepares the adaptive system to carry out a more robust immune response.

HOW PATHOGENS OVERCOME HOST DEFENCE MECHANISMS

By reading this article thus far, you may feel that humans may come on top of the arms race between humans and pathogens. Think again. Pathogens continually evolve new mechanisms to evade and overcome the human body's immune response. Similar to bacteria evolving to become resistant to antibiotics, pathogens evolve in ways to evade the immune system responses. Let's look at some of them.

Antigenic variation

Infectious disease-carrying pathogens can evade the immune surveillance of our bodies by altering its antigens, especially in extracellular pathogens. The principal defence of our bodies for this kind of invasion is the production of antibodies. These antibodies, in most cases, are not produced by the body when antigenic variation occurs since the pathogen alters the proteins or carbohydrates on its surface and avoids any sort of host immune response. DNA rearrangements in the pathogens account for the success of two prominent bacterial pathogens - *Salmonella typhimurium* and *Neisseria gonorrhoeae*.

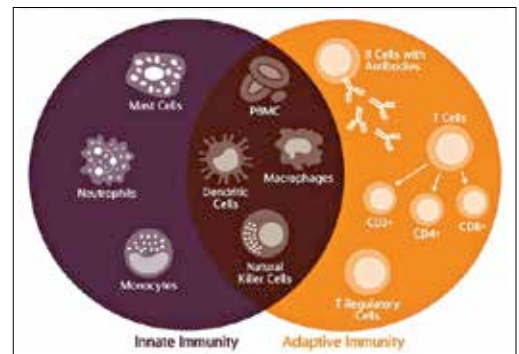
Persistence in vivo

Viruses also disclose their presence to the immune system once they have successfully entered cells by directing the synthesis of viral proteins. Fragments of this are displayed on the surface MHC molecules of the cell which has been infected, where T lymphocytes are detected. A virus, generally, in response, makes viral proteins and swiftly replicates viruses that produce acute viral diseases before they are detected by T

cells. However, some pathogens instead enter a state called latency where they do not replicate at all. They simply reside in the human system until the threat is dealt with and they can be reactivated and cause illnesses later.

Resisting destruction altogether

Some pathogens do induce regular immune response but have evolved special mechanisms to resist their effects. For example, *Mycobacterium tuberculosis* is taken up by macrophages but doesn't allow the fusion of phagosome with the lysosome, therefore shielding itself from bactericidal actions of all the lysosomal contents.



Innate and adaptive immune system

CONCLUSION

The arms race between man and pathogens is constantly tipping over one edge or the other owing to either pathogen developing more resistance towards the immune system or the humans coming up with new discoveries that hold the potential to stop disease-causing pathogen right in their tracks. Humans are constantly evolving medicine and cures, especially since the advent of technology, but we are yet to discover hundreds, if not thousands, of pathogens. Approximately 1400 human pathogens have been described to-date, however, it has been estimated that there are about one trillion microbial species on Earth, the vast majority of which are still uncharacterised. We are yet to discover all of these species and many of these may actually be human pathogens that will bring with them their own share of threats and challenges. ❏